

MQST Ensemble QC Laboratory

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Lab 4 - Quantum Algorithms

Winter quarter 2025

1 Introduction

In our final set of labs you will implement and run the two-qubit versions of the Deutsch-Jozsa and simulate the ground state energy of H_2 . This *should* simply be putting together the variety of things you have already done in both this lab and in the quantum algorithms class in the Fall. Specifically this will involve coding and running a basic quantum circuit using a pseudo-pure initial state, implementing a set of quantum oracles, and mapping quantum Hamiltonians.

1.1 Assigned Reading

- “Quantum Information Processing with NMR” : MIT-QCwithNMRlab.pdf.
- J. A. Jones, Quantum Computing with NMR, Progress in Nuclear Magnetic Resonance Spectroscopy 59, 91 (2011).
This is an excellent review of much of what we are doing. Sec. 2 is a brief review of what you have already learned. Sec’s 3 and 4 are highly relevant to this lab.
- Du J, Xu N, Peng X, Wang P, Wu S, Lu D. NMR Implementation of a Molecular Hydrogen Quantum Simulation with Adiabatic State Preparation. Physical review letters. 2010 Jan 22;104(3):030502.

Additional references from Lab 3 :

- E. Knill, I. Chuang, and R. Laflamme, Effective Pure States for Bulk Quantum Computation, Phys. Rev. A 57, 3348 (1998).
- Lu, D., Brodutch, A., Park, J., Katiyar, H., Jochym-O’Connor, T. and Laflamme, R., (2016). NMR quantum information processing. From Electron Spin Resonance (ESR) Based Quantum Computing, pp.193-226.
- D. G. Cory, A. F. Fahmy, and T. F. Havel, Ensemble Quantum Computing by NMR Spectroscopy, PNAS 94, 1634 (1997).
- N. A. Gershenfeld and I. L. Chuang, Bulk Spin-Resonance Quantum Computation, Science 275, 350 (1997).
- D. G. Cory, M. D. Price, and T. F. Havel, Nuclear Magnetic Resonance Spectroscopy: An Experimentally Accessible Paradigm for Quantum Computing, Physica D: Nonlinear Phenomena 120, 82 (1998).
- Aspuru-Guzik A, Dutoi AD, Love PJ, Head-Gordon M. Simulated quantum computation of molecular energies. Science. 2005 Sep 9;309(5741):1704-7.

2 Prelab

Please refer to the assigned reading and complete the following problems :

1. NMR gate set
 - i) Show how to implement the following gates in NMR with chloroform. I.e. write out the complete sequence of steps in terms of product operators and indicate what calibrated elemental gates you will need. Note : this should really just be writing down what you have done previously.

- (a) One-qubit X
 - (b) One-qubit Hadamard
 - (c) Two-qubit controlled-Z
 - (d) Two-qubit CNOT
- ii) Implement all of the above gates in Python/QuTiP in flexible and reusable fashion. These should form the core of the algorithm implementations you will write below in questions 3 and 4.
2. Preparatory Question 4 in “Quantum Information Processing with NMR”. See the file `MIT-QCwithNMRlab.pdf`.
 Note : If you are tempted to write out some 4×4 matrices in the first part, Don't! The quantum circuit representation should be evident. You will do this in the third part as verification.
 3. Two-qubit Deutsch-Jozsa : Write out the full procedure for implementing the two-qubit Deutsch-Jozsa algorithm in terms of our available NMR gates. This must include the 'state-preparation' of $|0\rangle|0\rangle$ and subsequent accumulation of the three associated measurements. Implement and demonstrate correctness using QuTiP.
 Note : You have just built up the four required Oracle functions in the previous problem. You have also built up the required pseudo-state preparation circuits in Lab 3. Now put it all together.
 IMPORTANT : Your group **must** complete this problem before coming to the lab. Otherwise your lab time will be completely wasted.
 4. Quantum Simulation of ground-state energy of H_2 . See the file “NMR Implementation of a Molecular Hydrogen Quantum Simulation with ASP.pdf”.
Adiabatic State Preparation (ASP): Explain how ASP is used to prepare the system qubit in the ground state of the H_2 molecular Hamiltonian. What is the role of the initial Hamiltonian H_0 (e.g., $-\sigma_x$) and how does the linear interpolation to the target Hamiltonian H ensure that the qubit remains in the ground state?
Iterative Phase Estimation: Describe the iterative phase estimation procedure employed in the experiment. How is the controlled- U_k operation implemented, and how do successive iterations (with feedback from previous phase measurements) lead to the extraction of the molecular energy?

3 Lab Procedure

Read the material and follow the instructions in the Python notebook *NMR_Lab4.ipynb* found on BruinLearn and on the Magritek systems.